

If statement Diagnostic test: SOLUTIONS

The results do not count for anything. This test is to allow the ECS401 team to diagnose if you need help to ensure they do not fall behind in learning to program. Understanding how if statements works is critical so if you have any problems we want to sort them out now.

You must show all your working in the following. No correct table (exactly the same)... no marks.

1. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
if (a == 10)  // L3
    {a = 11;} // L4
else
    {b = 11;} // L5
```

LINE	a	b	a==10?
L1	10		
L2		20	
L3			10==10? true
L4	11		

a is now 11

b is now 20

2. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
if (a == 11)  // L3
    {a = 11;} // L4
else
    {b = 11;} // L5
```

LINE	a	b	a==11?
L1	10		
L2		20	
L3			10==11? false
L5		11	

a is now 10

b is now 11

3. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
if (a == 11)  // L3
    {a = 12;} // L4
else
    {b = 11;} // L5
a = 13;      // L6
```

LINE	a	b	a==11?
L1	10		
L2		20	
L3			10==11? false
L5		11	
L6	13		

a is now 13

b is now 11

4. What are the new values of a and b after the following code fragment has executed and what is printed?

```
String a = "5";           // L1
String b = "6";           // L2
if (a.equals("b"))        // L3
    {System.out.println(a);} // L4
else
    {System.out.println(b);} // L5
System.out.println("DONE"); // L6
```

a is now ____ "5" ____ b is now ____ "6" ____

Printed

6
DONE

LINE	a	b	a.equals("b")?	Printed
L1	"5"			
L2		"6"		
L3			"5".equals("b")? false	
L5				6
L6				DONE

5. What are the new values of a and b after the following code fragment has executed and what is printed?

```
String a = "5";           // L1
String b = "6";           // L2
if (!(a.equals("b")))      // L3
    {System.out.println(a);} // L4
else
    {System.out.println(b);} // L5
System.out.println("DONE"); // L6
```

a is now ____ "5" ____ b is now ____ "6" ____

Printed

5
DONE

LINE	a	b	!(a.equals("b"))?	Printed
L1	"5"			
L2		"6"		
L3			!("5".equals("b"))? true	
L4				5
L6				DONE

6. What are the new values of a, b and c after the following code fragment has executed?

```
int a = 5;                // L1
int b = 6;                // L2
if (a == b)               // L3
    {a = 8;}              // L4
else if (a > b)            // L5
    {a = 9;}              // L6
else                      // L7
    {a = 10;}             // L7
```

a is now ____ 10 ____ b is now ____ 6 ____

LINE	a	b	a==b?	a>b?
L1	5			
L2		6		
L3			5==6? false	
L5				5>6? false
L7	10			

7. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;           // L1
int  b = 6;           // L2
if (a < b)             // L3
    {a = 8;}          // L4
else if (a > b)        // L5
    {a = 9;}          // L6
else                  // L7
    {a = 10;}
```

a is now 8 **b is now** 6

LINE	a	b	a<b?	a>b?
L1	5			
L2		6		
L3			5<6? true	
L4	8			

8. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;           // L1
int  b = 6;           // L2
if (a >= b)            // L3
    {a = 8;}          // L4
else if (a < b)        // L5
    {a = 9;}          // L6
else                  // L7
    {a = 10;}
```

a is now 9 **b is now** 6

LINE	a	b	a>=b?	a<b?
L1	5			
L2		6		
L3			5>=6? false	
L5				5<6? true
L6	9			